

Topic 02: Slope Fields, Solution Curves & Qualitative Behavior

Professor Leonard Master Series · Lessons 5 & 6

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: Visualizing the Wind

"Imagine you are looking down at an ocean current from an airplane. At every single point (x, y) in the water, an arrow points in the direction the current is flowing."

That is precisely what a **Slope Field (Direction Field)** is! Given $\frac{dy}{dx} = f(x, y)$:

- You do **not** need to integrate.
- You do **not** need to find an algebraic formula.
- At every point (x, y) , calculate the number $f(x, y)$. That number is the tangent slope m .
- Draw a tiny line segment with slope m .

If you drop a boat (an initial condition (x_0, y_0)) into the ocean, the boat will follow the current. The path the boat traces out is the **solution curve**!

2. Curriculum Reference Diagram

consequently it may be possible to see at a glance certain qualitative aspects of the solutions—regions in the plane, for example, in which a solution exhibits an unusual behavior. A single solution curve that passes through a direction field must follow the flow pattern of the field; it is tangent to a lineal element when it intersects a point in the grid.

EXAMPLE 1 Direction Field

The direction field for the differential equation $dy/dx = 0.2xy$ shown in **FIGURE 2.1.2(a)** was obtained using computer software in which a 5×5 grid of points (mh, nh) , m and n integers, was defined by letting $-5 \leq m \leq 5$, $-5 \leq n \leq 5$ and $h = 1$. Notice in Figure 2.1.2(a) that at any point along the x -axis ($y = 0$) and the y -axis ($x = 0$) the slopes

Key Observations:

1. Along the axes where $x = 0$ or $y = 0$, the slope is $0.2(0) = 0$, so all hash marks are **horizontal**.
2. In Quadrant I ($x > 0, y > 0$), slopes are positive and get steeper as x and y grow.
3. Solution curves entering Quadrant I bend upwards exponentially!

3. Isoclines: The Fast Way to Sketch by Hand

Calculating slopes point-by-point takes forever. Professor Leonard uses **isoclines** (curves of constant slope): Set $f(x, y) = c$, where c is a constant slope.

- For $\frac{dy}{dx} = x - y$:
 - Set slope $c = 0 \implies x - y = 0 \implies y = x$. Everywhere along the line $y = x$, draw horizontal tick marks ($m = 0$)!
 - Set slope $c = 1 \implies x - y = 1 \implies y = x - 1$. Everywhere along $y = x - 1$, draw 45° tick marks ($m = 1$).
 - Set slope $c = -1 \implies y = x + 1$. Draw tick marks with slope -1 .

Now connect the flow: any curve crossing $y = x$ has a horizontal tangent (a local minimum or maximum)!

4. Leonard's Red Flag Alerts

-  **Red Flag:** Drawing solution curves that cross each other. If $f(x, y)$ and $\partial f / \partial y$ are smooth, **two solution curves can NEVER intersect**. If they crossed, a boat placed at the intersection wouldn't know which way to flow!